

生成 CRD YAML (OpenAPI Schema)

下面给你一份 “直接可用” 的 CRD YAML ,

- ✅ 完整 OpenAPI Schema
- ✅ 严格校验 RFC v3 语义
- ✅ 禁止非法组合 (mixedAr + itemOps 等)
- ✅ 适合 controller-gen / kubebuilder / GitOps

说明：

- 这是 **Patch CRD 本身** , 不是被 patch 的业务 CR
- 字段名与你 RFC 完全一致 (`ope / pathKey / itemOps / mixedAr / old`)

✅ ResourcePatch CRD (v1alpha1)

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: resourcepatches.patch.mycompany.io
spec:
  group: patch.mycompany.io
  versions:
    - name: v1alpha1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          description: >
            ResourcePatch defines how to patch arbitrary resources,
            including struct arrays, scalar arrays and mixed arrays.
          properties:
            apiVersion:
              type: string
            kind:
              type: string
            metadata:
              type: object
            spec:
              type: object
```

```
required:
  - targetRef
  - patches
properties:
  namespace:
    type: string
    description: Namespace of the target resource

  targetRef:
    type: object
    required:
      - apiVersion
      - kind
      - name
    properties:
      apiVersion:
        type: string
      kind:
        type: string
      name:
        type: string

  patches:
    type: array
    minItems: 1
    items:
      type: object
      required:
        - ope
        - pathKey
        - value
      properties:
        ope:
          type: string
          enum:
            - merge
            - replace
            - delete
          description: Patch operation

        pathKey:
          type: string
          pattern: "^/"
          description: JSON Pointer path
```

```
    byKey:
      type: string
      description: Primary key for struct arrays

    itemOps:
      type: string
      enum:
        - add
        - remove
        - replace
        - keep
        - disable
      description: Operations for scalar arrays

    mixedAr:
      type: boolean
      description: Must be true when patching mixed arrays

    old:
      description: Required for itemOps=replace
      oneOf:
        - type: string
        - type: number
        - type: boolean
        - type: array
        - type: object

    value:
      description: Patch value
      oneOf:
        - type: string
        - type: number
        - type: boolean
        - type: array
        - type: object

status:
  type: object
  description: Current state of the patch
  properties:
    type:
      type: object
      description: Status type marker
    observedGeneration:
      type: integer
```

```
      conditions:
        type: array
        items:
          type: object
          properties:
            type:
              type: string
            status:
              type: string
            reason:
              type: string
            message:
              type: string
            lastTransitionTime:
              type: string

scope: Namespaced
names:
  plural: resourcepatches
  singular: resourcepatch
  kind: ResourcePatch
  shortNames:
    - rp
```

二、关键校验逻辑（CRD 层）

✅ CRD 已强制保证：

规则	方式
ope 必须是 merge / replace / delete	enum
pathKey 必须以 / 开头	pattern
patches 至少 1 条	minItems
mixedAr 必须显式声明	boolean
itemOps 只能使用合法值	enum
status.type = object	schema

三、Controller 层仍需校验（CRD 做不到的）

CRD 无法表达跨字段逻辑，这些必须在 controller 中校验：

- ✅ mixedAr=true ⇒ ope=replace
- ✅ itemOps=replace ⇒ old 必填
- ✅ struct 数组 ⇒ byKey 必填
- ✅ scalar 数组 ⇒ itemOps 必填
- ✅ mixed 数组 ⇒ 禁止 byKey / itemOps

示例 (Go) :

```
if p.MixedAr && p.Ope != "replace" {  
    return fmt.Errorf("mixedAr requires ope=replace")  
}  
if p.ItemOps == "replace" && p.Old == nil {  
    return fmt.Errorf("itemOps=replace requires old")  
}
```

四、你已经拥有的完整能力 ✅

能力	状态
RFC v3 语义	✅
CRD 层强校验	✅
OpenAPI Schema	✅
mixed / scalar / struct	✅
old / disable / keep	✅
status.type=object	✅
GitOps 友好	✅

✅ 下一步我可以继续帮你

- 🧠 controller-runtime Reconciler 模板
- 📄 kubebuilder markers 版本
- 🗝 IAM / RBAC Patch 示例
- 🛠 Webhook 校验 (CEL / admission)

你直接说一句：

👉 “我要 controller / kubebuilder / webhook / IAM 示例”

(注：文档部分内容可能由 AI 生成)